

Generative AI Complete Notes with Theory and Answers

Introduction to Generative AI

Generative AI is a branch of artificial intelligence that focuses on creating new content such as text, images, audio, video, and code. It uses machine learning and deep learning models to generate outputs similar to human-created content.

Questions and Answers

Q: What is Generative AI?

Answer: Generative AI is a technology that creates new content using machine learning and deep learning techniques.

Q: What are the applications of Generative AI?

Answer: Generative AI is used in chatbots, image generation, coding assistants, healthcare, education, and content creation.

Machine Learning Basics

Machine Learning is a subset of AI where systems learn patterns from data without explicit programming.

Questions and Answers

Q: What is Machine Learning?

Answer: Machine Learning is a technique where computers learn patterns from data and improve automatically.

Q: What are the types of Machine Learning?

Answer: Supervised learning, unsupervised learning, and reinforcement learning.

Deep Learning

Deep Learning is a subset of machine learning that uses artificial neural networks with multiple hidden layers.

Questions and Answers

Q: What is Deep Learning?

Answer: Deep Learning uses neural networks to process complex data and solve advanced AI problems.

Q: What are neural networks?

Answer: Neural networks are systems inspired by the human brain used for learning patterns.

Natural Language Processing

NLP allows computers to understand, process, and generate human language.

Questions and Answers

Q: What is NLP?

Answer: NLP stands for Natural Language Processing and helps machines understand human language.

Q: What are the applications of NLP?

Answer: Chatbots, translation systems, sentiment analysis, and virtual assistants.

Large Language Models

Large Language Models are AI models trained on massive datasets to understand and generate text.

Questions and Answers

Q: What is an LLM?

Answer: LLM stands for Large Language Model used for text generation and understanding.

Q: Name some popular LLMs.

Answer: GPT, Gemini, Claude, and Llama.

Transformers

Transformers are deep learning architectures widely used in Generative AI.

Questions and Answers

Q: What is a transformer model?

Answer: A transformer is a neural network architecture used in NLP and Generative AI.

Q: Why are transformers important?

Answer: They process large amounts of data efficiently and improve language understanding.

Prompt Engineering

Prompt engineering is the process of designing effective prompts to get accurate AI responses.

Questions and Answers

Q: What is prompt engineering?

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Q: Why are prompts important?

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Applications of Generative AI

Generative AI is transforming industries through automation and intelligent systems.

Questions and Answers

Q: Where is Generative AI used?

Answer: Healthcare, education, finance, gaming, and software development.

Q: How does Generative AI help developers?

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